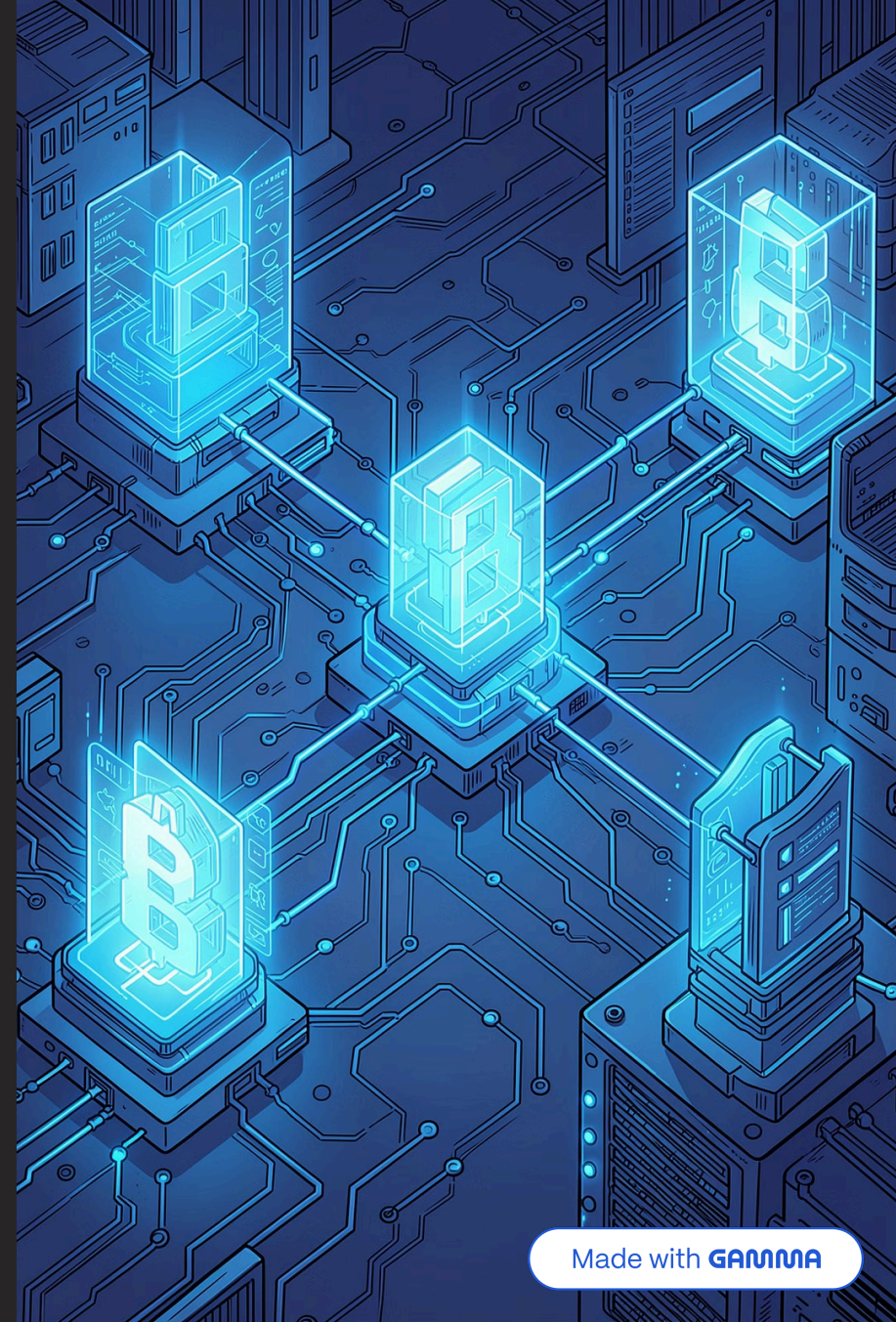


Blockchain-Based Academic Certificate Verification System

MCA Major Project-II (CA401P) · Session 2025–2026

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An illustration of a desk with a clipboard, a pen, and a red circular stamp. The clipboard has a document titled 'ABGDDORQUMENT CERTIEICAATE' and 'FOGENIC TRUTUK FRND'. The document contains several lines of text, some of which are crossed out. A hand is holding a pen and writing on the document. A red circular stamp is visible on the document, with the text 'MOAN' and 'S. 13' inside it. The background shows a desk with books and a pen holder.

The Problem: Academic Credential Fraud

Traditional certificate verification is slow, costly, and vulnerable to manipulation — creating a crisis of trust across institutions and employers.

Certificate Forgery

Counterfeit credentials are easy to produce and difficult to detect with manual processes.

Manual Delays

Verification can take weeks, delaying hiring and admissions decisions.

No Interoperability

Institutions lack a shared, trusted system for cross-verification of credentials.

High Admin Cost

Administrative overhead for issuing and verifying certificates is substantial.

Project Objectives



- **Prevent Certificate Fraud**
Use cryptographic hashing to make certificates tamper-evident.
- **Automate Verification**
Smart contracts eliminate manual checks and reduce human error.
- **Ensure Transparency & Security**
All records are immutable and publicly auditable on the blockchain.
- **Enable Real-Time Validation**
Employers verify credentials instantly via QR code lookup.

Why Blockchain?

Blockchain's core properties directly address every gap in traditional certificate verification systems.



Decentralization

No single point of failure or control — distributed across the network.



Immutability

Once written, records cannot be altered or deleted.



Transparency

All transactions are visible and auditable by authorized parties.



Security

Cryptographic techniques protect data integrity at every layer.



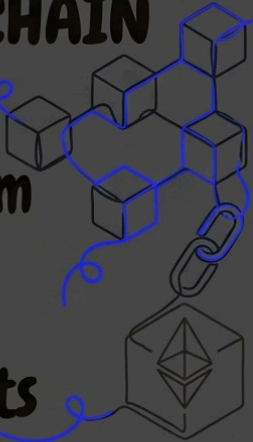
Trustless Verification

Parties verify without needing to trust a central authority.

Technology Stack

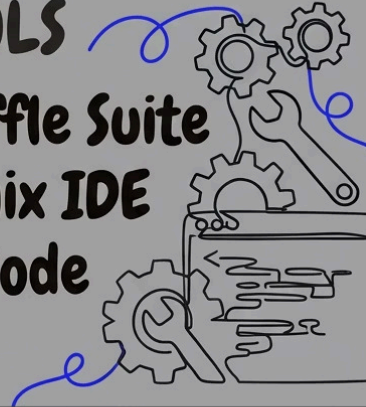
BLOCKCHAIN LAYER

Ethereum
Solidity
Smart
Contracts



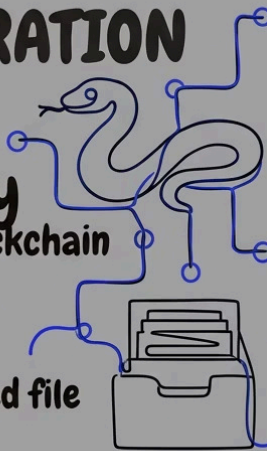
DEVELOPMENT TOOLS

Truffle Suite
Remix IDE
VS Code
Git



INTEGRATION LAYER

Web3.py
Python-Blockchain
interaction
IPFS
Decentralized file
storage



SECURITY LAYER

SHA-256
Hashing
QR Code
Generation



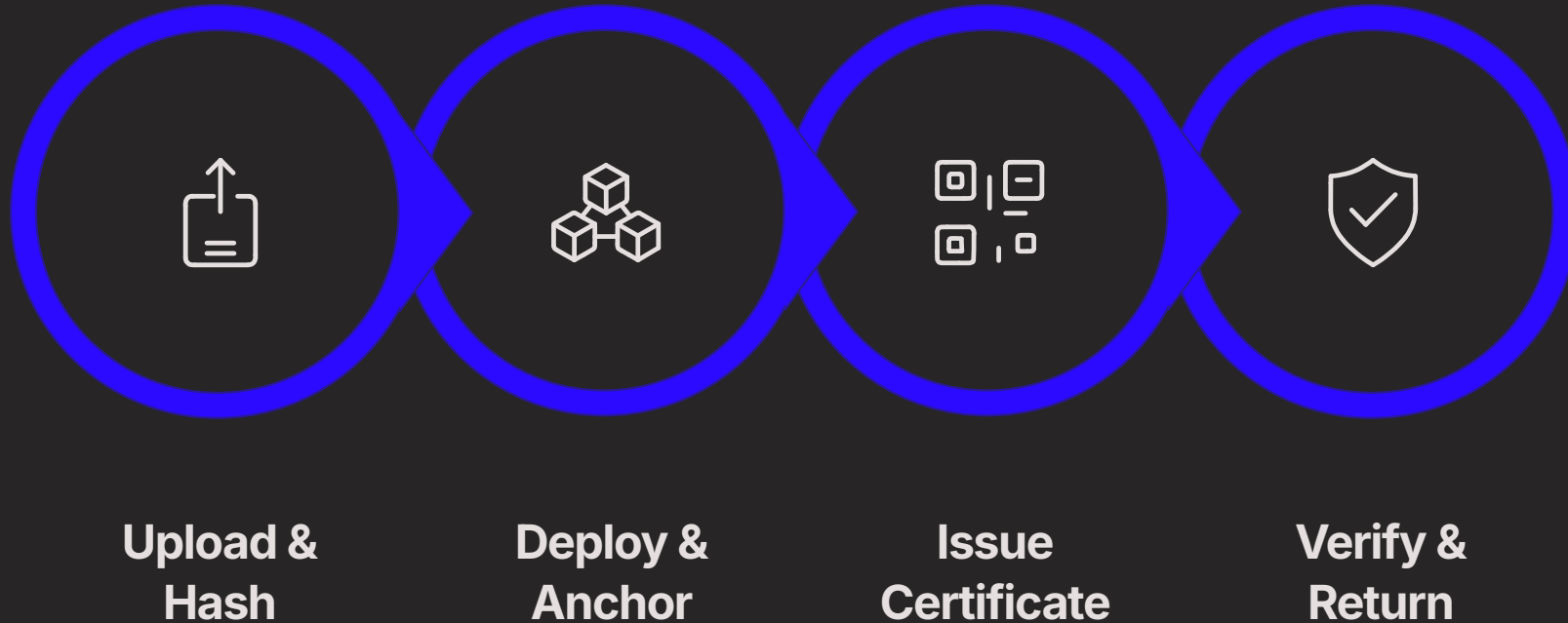
Core Platform

Ethereum provides the decentralized ledger; **Solidity** powers smart contracts for issuance, verification, and revocation.

Development & Storage

Truffle Suite and **Remix IDE** streamline contract deployment. **IPFS** stores certificate metadata off-chain. **Web3.py** bridges Python backend with the blockchain.

System Architecture & Workflow



The end-to-end pipeline ensures that every certificate passes through cryptographic verification before being permanently recorded on the blockchain — making tampering computationally infeasible.

Cryptographic Hashing & Smart Contracts

SHA-256 Hashing

Each certificate is passed through SHA-256, producing a unique 256-bit fingerprint. Any change to the original document — even a single character — produces a completely different hash, enabling instant tamper detection.

Key Property: SHA-256 is deterministic, one-way, and collision-resistant — ideal for credential integrity.

Smart Contracts

Issuance Contract

Institution registers and issues certificates on-chain.

Verification Contract

Compares submitted hash against stored record automatically.

Revocation Contract

Allows institutions to flag compromised credentials.

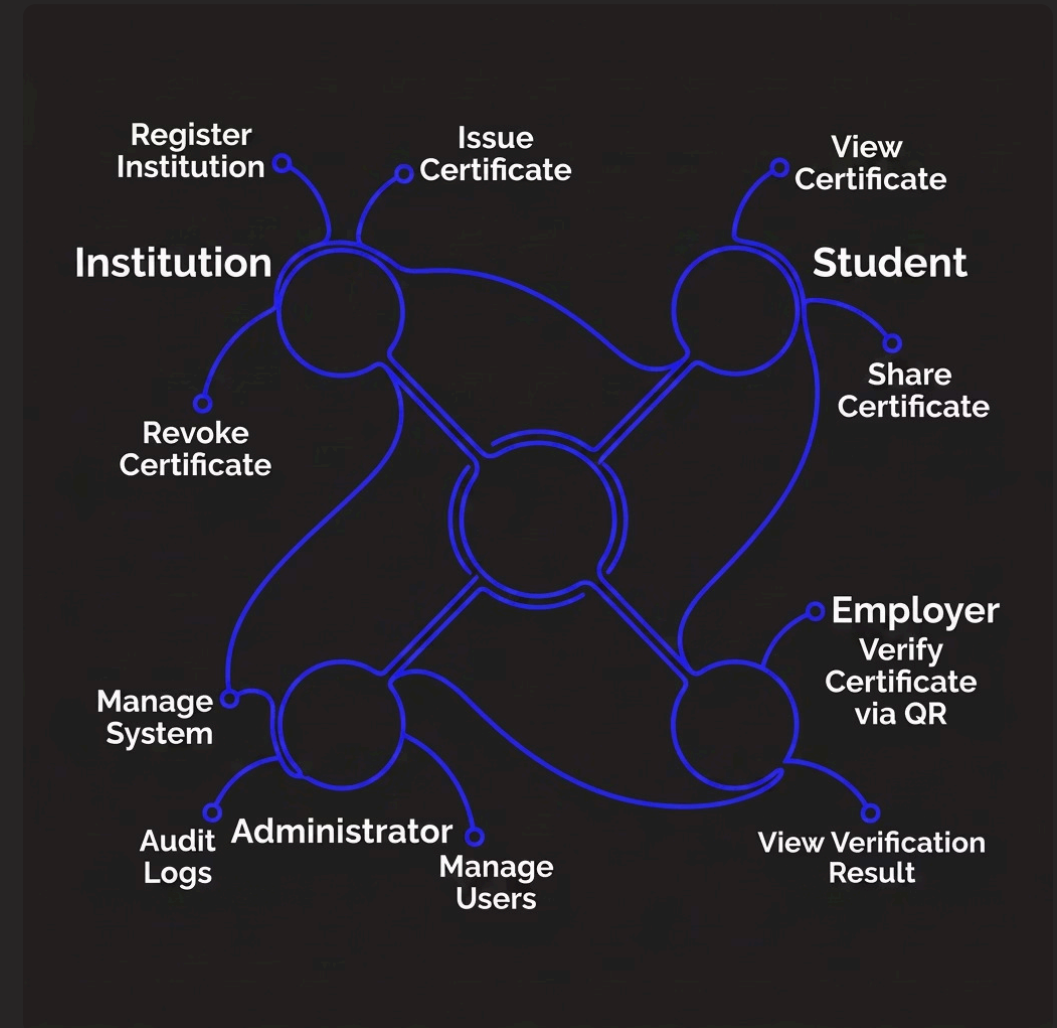
System Design: DFD, ER & Use Cases

Data Flow & Entity Model

Level-0 DFD: Institution → System → Student/Employer

Level-1 DFD: Upload → Hash → Store → Verify → Result

Key ER Entities: Institution, Student, Course, Certificate, Smart Contract, Verifier, Verification Log, Admin, Audit Log





Features, Advantages & Limitations

Core Features

- Secure multi-role authentication
- Blockchain-backed verification
- Immutable audit trails
- Certificate lifecycle management
- Real-time QR validation

Advantages

- Tamper-proof, forgery-resistant records
- Instant verification — seconds vs. weeks
- Full transparency for all stakeholders
- Significantly lower operational costs

Limitations

- Scalability constraints on Ethereum
- Smart contract vulnerability risks
- Computational and gas cost overhead
- Complex implementation and onboarding

Future Scope & Conclusion

What's Next

01

Multi-University Integration

Expand to a consortium of institutions on a shared ledger.

02

Mobile Application

Student-facing app for certificate access and sharing.

03

AI-Assisted Fraud Detection

ML models to flag anomalous issuance patterns.

04

Government Credential Integration

Link with national digital credential frameworks.

Conclusion

This system demonstrates that blockchain technology can effectively eliminate academic certificate fraud, replacing slow manual processes with instant, trustless verification.

It enhances trust among institutions, students, and employers — supporting the broader digital transformation of education.

Thank You

Questions & Answers